Availability & Reliability

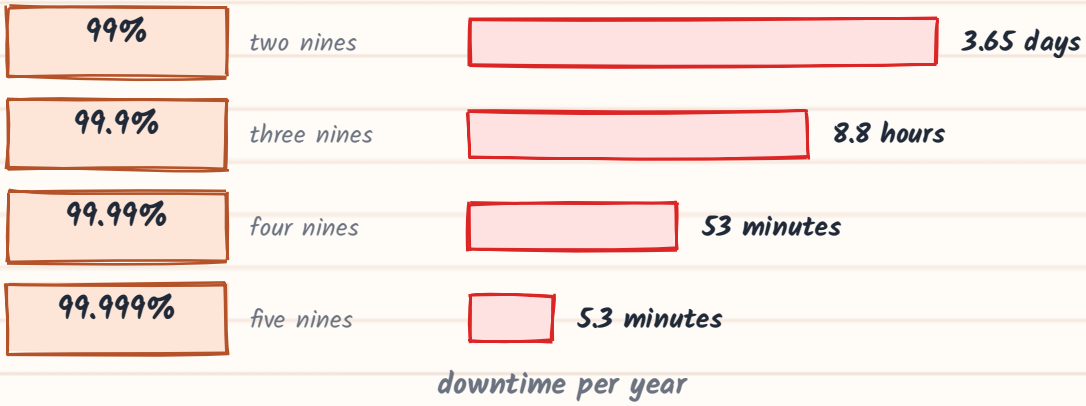
Lesson 4 · SLAs, the nines, redundancy & failover

In this lesson → ① SnapLink goes down ② Availability vs reliability ③ The nines ④ SLI · SLO · SLA ⑤ Series vs parallel ⑥ SPOFs ⑦ Failover ⑧ More HA tools ⑨ Interview Q&A
Running project: SnapLink, our photo-sharing app that grows from 1 server to 100M users.

1 Availability vs reliability

Availability = % of time the system is up and responding. **Reliability** = keeps working correctly over time (no lost data, no wrong answers).

2 The nines



Availability	Downtime / year	Downtime / month
99%	~3.65 days	~7.3 hours
99.9%	~8.8 hours	~44 minutes
99.99%	~53 minutes	~4.4 minutes
99.999%	~5.3 minutes	~26 seconds

3 SLI · SLO · SLA

Term	Meaning	Example
SLI (indicator)	what you measure	% successful requests, p99 latency
SLO (objective)	internal target	99.95% success per month
SLA (agreement)	contract with penalty	99.9% or service credits

> **Rule**
> Set the SLA **looser** than the SLO so you have a safety margin (error budget).

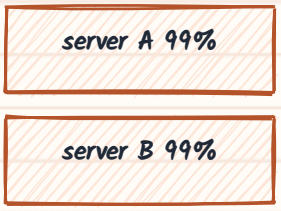
4 Series vs parallel

in series: all must work



$= 99.9\% \times 99.9\% \times 99.9\% \approx 99.7\%$

in parallel: one is enough

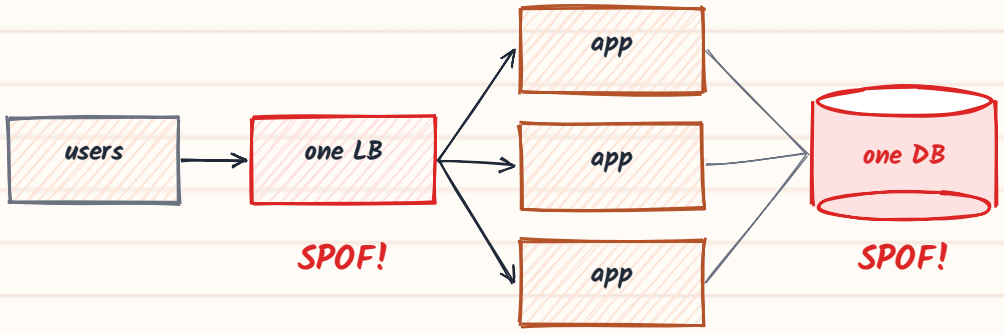


$= 1 - (1\%)^2 = 99.99\%$

→ Series: $A = A1 \times A2 \times \ldots$ (goes **down**)

→ Parallel: $A = 1 - (1 - A1)(1 - A2)\ldots$ (goes **up**)

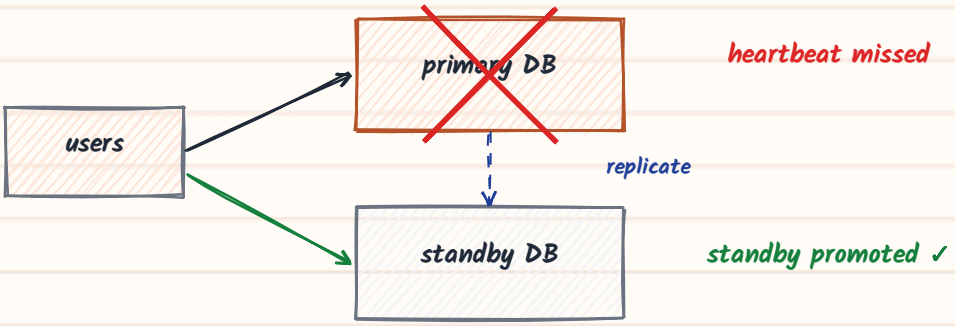
5 Single points of failure



fix: 2+ of everything, in 2+ zones

- ✓ At least 2 of every component
- ✓ Spread across 2+ availability zones
- ✓ Remove hidden SPOFs: DNS, config, one admin, one region

6 Failover



Mode	How	Trade-off
Active-passive	standby takes over on failure	simple; standby idle; brief failover gap
Active-active	all nodes serve traffic	more capacity; harder consistency

7 More high-availability tools

- ✓ Health checks (LB removes bad servers)
- ✓ Graceful degradation (feed works without recommendations)
- ✓ Timeouts, retries, circuit breakers
- ✓ Backups + tested restores
- ✓ Multi-region for disasters (SD-23)

Interview tip
Say where each **SPOF** is and how you remove it; that's what "high availability" means in a design.

8 Interview Corner

Q. What does 99.99% availability mean?

A. About 53 minutes of downtime per year (~4.4 min/month).

Q. Availability vs reliability?

A. Up and responding vs working correctly over time.

Q. How does redundancy change availability?

A. Parallel components multiply unavailability: two 99% servers → 99.99%.

Q. How do you remove SPOFs?

A. Duplicate every component across zones with health checks and automatic failover.

Q. SLO vs SLA?

A. SLO is the internal target; SLA is the contractual promise with penalties.

⚡ Revise in 60 seconds

- → 99.9% ≈ **8.8 h**/yr
- → 99.99% ≈ **53 min**/yr
- → 99.999% ≈ **5 min**/yr
- → SLI → SLO → SLA
- → Series ↓, parallel ↑
- → No **SPOF**: 2 of everything
- → 2+ zones
- → Automatic **failover**

Next lesson → SD-05
CAP, PACELC & consistency models.
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**